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Experiment: 6

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6. Gradient Descent, SGD, Momentum and Adam Comparison

Scenario

A linear regression model predicts house prices but initially produces high prediction error. Implement gradient descent and visualize how the loss changes over successive iterations**.**

Algorithm

1. Start the program.
2. Import NumPy, TensorFlow/Keras, Pandas, and Matplotlib libraries.
3. Create a dataset containing house sizes and their corresponding prices.
4. Normalize the input features to improve training.
5. Build a linear regression model using TensorFlow/Keras.
6. Compile the model using Mean Squared Error (MSE) as the loss function and Gradient Descent (SGD) as the optimizer.
7. Train the model for a specified number of iterations.
8. Record the loss value at each iteration.
9. Display a summary table of the loss values.
10. Plot the loss against the number of iterations.
11. Display actual and predicted house prices.
12. Stop the program.

Source code

```
import numpy as np

import tensorflow as tf

import matplotlib.pyplot as plt

from tensorflow import keras

from tensorflow.keras import layers
```

```
# 1. Create the dataset
# House size in square feet
X = np.array([[800], [1000], [1200], [1500], [1800],
              [2000], [2200], [2500]], dtype=float)

# House price in lakhs
y = np.array([[25], [30], [35], [42], [50],
              [55], [60], [68]], dtype=float)

# Normalize input features
X_scaled = X / 1000.0

# 2. Build the linear regression model
model = keras.Sequential([
    layers.Input(shape=(1,)),
    layers.Dense(1, activation="linear")
])

# 3. Compile using Gradient Descent (SGD)
model.compile(
    optimizer=keras.optimizers.SGD(learning_rate=0.01),
    loss="mean_squared_error"
)

# 4. Record initial loss
initial_loss = model.evaluate(X_scaled, y, verbose=0)

# Store the loss at every epoch
```

```
loss_values = []

class LossHistory(keras.callbacks.Callback):
    def on_epoch_end(self, epoch, logs=None):
        loss_values.append(logs["loss"])

# 5. Train the model
history = LossHistory()

model.fit(
    X_scaled, y,
    epochs=200,
    batch_size=len(X),
    verbose=0,
    callbacks=[history]
)

# 6. Predict house prices
predictions = model.predict(X_scaled, verbose=0)

# 7. Display model summary
print("\nMODEL SUMMARY")
model.summary()

# 8. Display loss summary table
print("\nGRADIENT DESCENT LOSS SUMMARY")
print("-" * 48)
print(f"{'Iteration':<15}{'Loss (MSE)':>20}")
print("-" * 48)
```

```
print(f"{'Initial':<15}{initial_loss:>20.4f}")

for i in range(0, len(loss_values), 20):
    print(f"{i + 1:<15}{loss_values[i]:>20.4f}")

print(f"{len(loss_values):<15}{loss_values[-1]:>20.4f}")
print("-" * 48)

# 9. Display actual and predicted prices
print("\nHOUSE PRICE PREDICTION TABLE")
print("-" * 52)
print(f"{'Size (sq.ft.)':<18}{'Actual Price':>16}{'Predicted Price':>18}")
print("-" * 52)

for i in range(len(X)):
    print(f"{X[i, 0]:<18.0f}{y[i, 0]:>13.2f} L"
          f"{predictions[i, 0]:>14.2f} L")

print("-" * 52)
print(f"\nInitial Loss: {initial_loss:.4f}")
print(f"Final Loss:   {loss_values[-1]:.4f}")

# 10. Plot loss vs iterations
plt.figure(figsize=(9, 5))
plt.plot(
    range(1, len(loss_values) + 1),
    loss_values,
    label="Training Loss",
    linewidth=2
```

```
)
plt.xlabel("Iterations (Epochs)")
plt.ylabel("Mean Squared Error (MSE)")
plt.title("Gradient Descent: Loss vs Iterations")
plt.grid(True)
plt.legend()
plt.tight_layout()
plt.show()

# 11. Plot actual vs predicted house prices
plt.figure(figsize=(9, 5))
plt.plot(X[:, 0], y[:, 0], "o-", label="Actual Prices")
plt.plot(X[:, 0], predictions[:, 0], "s--",
        label="Predicted Prices")
plt.xlabel("House Size (sq.ft.)")
plt.ylabel("House Price (Lakhs)")
plt.title("Actual vs Predicted House Prices")
plt.grid(True)
plt.legend()
plt.tight_layout()
plt.show()
```

Output

Model: "sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 1)	2

Total params: 4 (20.00 B)

Trainable params: 2 (8.00 B)

Non-trainable params: 0 (0.00 B)

Optimizer params: 2 (12.00 B)

GRADIENT DESCENT LOSS SUMMARY

Iteration	Loss (MSE)
Initial	2031.1555
1	2031.1556
21	84.7839
41	7.3222
61	4.0021
81	3.6382
101	3.4064
121	3.1935
141	2.9944
161	2.8078
181	2.6331
200	2.4773

HOUSE PRICE PREDICTION TABLE

Size (sq.ft.)	Actual Price	Predicted Price
800	25.00 L	27.43 L
1000	30.00 L	31.94 L
1200	35.00 L	36.44 L
1500	42.00 L	43.20 L
1800	50.00 L	49.95 L
2000	55.00 L	54.46 L
2200	60.00 L	58.96 L
2500	68.00 L	65.72 L

Initial Loss: 2031.1555
Final Loss: 2.4773

```
Initial Loss: 2031.1555
Final Loss:   2.4773
```



